

★ SPECIAL ISSUES

○ SPECIAL ISSUE 1

- When the instance is shut down the data is lost. To stop this from occurring we need to export the EC2 instance as a VM into an Amazon S3 bucket. The cost of this would be incredibly inexpensive. For anything less than a single GB it will cost about \$0.023 per month that is stored in S3 bucket. Creating the AMI itself is free, the cost only comes from the storage itself.
- The AWS docs are pretty vague about the process itself but from several linked documents I've gathered that you need to export your already existing and working EC2 instance (the AMI is already installed on EC2 instance) into a VM using VM import to Citrix Xen, Microsoft Hyper-V, or VMware vSphere. From there you need to create the S3 bucket where you wish to store it, make sure to pick the correct region and configure it correctly using your console or the AWS command line interface (CLI). After that is set up you need to VM export it into your S3 bucket with your S3 key using the AWS CLI. From there if you want to restore the instance you need to VM import from your S3 bucket and it'll bring the instance back as it was.
- We ran this through ChatGPT to check if our understanding of the issue was correct and it corrected a few parts. Here is what it had to say:
 - For normal EC2 backup and recovery, you should recommend:
 - Create an EBS snapshot for data backup.
 - Create an EBS-backed AMI to preserve the configured server.
 - Launch a replacement EC2 instance from the AMI if necessary.
 - Creating the AMI itself does not represent the main storage cost, but AWS charges for the EBS snapshots behind it.
 - VM Import/Export is mainly for moving an EC2 instance into an external virtualization environment such as VMware, Hyper-V, or Citrix Xen. It is not normally necessary just to protect an EC2 instance from shutdown. AWS also charges S3 storage and potentially temporary EBS snapshot costs for an export.
 - Your \$0.023 per GB-month statement should also say that this is a common S3 Standard rate in some regions. S3 prices vary by region and storage class, and there is no minimum storage charge.

○ SPECIAL ISSUE 2

- When you reboot or redeploy your EC2 instance you may have a different ip address. This is because there was not a specified private IP address chosen/set in the settings and one of the available addresses was assigned to it automatically. This can be remedied by specifying the ip

address when launching the instance, or at least specifying the range of addresses in your subnet.

- In an unprofessional setting this would be fine, but for professional work and actual deployment, you shouldn't have this happen. You can use an elastic IP address for this. An elastic IP address can be allocated to your AWS account and will incur charges whether you use it or not. This IP will not change over time and will be yours until you release it. The pricing is different for these. For our free tier AWS accounts we'll have 750 hours of use for free for the first 12 months. After that we will be charged roughly \$0.005 per hour of the instance running with an elastic IP address. The estimate is around \$3.60 per month. It's inexpensive but we only have around \$100 of free money to use before dipping into our own funds. This is of course for an elastic public IP that is reachable from the public, otherwise it'd be a huge waste of money.
- This can be done on the AWS Learner lab however the funds will be deducted from your account the same way, whether it is in use or idle. It'd be a good idea to release the elastic IP when you're no longer using it.
- We again ran this part through ChatGPT to check our understanding and here is what it said about this section:
 - An EC2 reboot is different from stopping and starting the instance.
 - An automatically assigned public IPv4 address can change after a stop/start cycle.
 - The private IPv4 address and network interface normally persist when the instance is stopped. However, terminating the instance and creating a new one can result in a different private IP. Amazon Elastic Compute Cloud
 - Specifying a subnet range does not guarantee one permanent address. You would need to assign a specific private IP or preserve the network interface.
 - An Elastic IP is static and is appropriate when the instance needs a permanent public IPv4 address. Amazon Elastic Compute Cloud
 - The current public IPv4 charge is \$0.005 per hour, or approximately \$3.65 per month for 730 hours. AWS charges both in-use and idle public IPv4 addresses. aws.amazon.com
 - The Free Tier wording should say "750 hours per month for the first 12 months for eligible accounts," not simply 750 total hours.